Exercise 1 — Step-by-Step Solutions

Optimization-Based Control Methods · Linear Systems and Linear Controllers

Course: Optimization-Based Control Methods (TU Chemnitz, Automatic Control Group, Dr. Adrian Redder) **Covers:** eigenvalue practice, Tasks 0–3 (all parts) and the numerical experiment **Reference:** checked against the official *Exercise Solution 1*; where the official solution argues differently, both routes are shown

Every step is written out. Section references like "Concepts §5.4" point to `01_Concepts.md`, which explains the theory behind each step. The Python scripts in `python/` verify every number.

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Part A — Eigenvalue practice

Method used throughout (Concepts §2.3): (1) form $\lambda I - A$; (2) take $\det(\lambda I - A)$; (3) find the roots; (4) check $\sum \lambda = \operatorname{tr} A$ and $\prod \lambda = \det A$.

Stability rule (Concepts §4.1): $x_{k+1} = Ax_k$ is asymptotically stable $\iff$ every eigenvalue has $|\lambda| < 1$.

Problem 1

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1. $\lambda I - A = \begin{bmatrix} \lambda - 4 & -1 \\ -2 & \lambda - 3 \end{bmatrix}$.

Step 2. $\det(\lambda I - A) = (\lambda - 4)(\lambda - 3) - (-1)(-2) = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10$.

Shortcut check: $\operatorname{tr} A = 7$ and $\det A = 4 \cdot 3 - 1 \cdot 2 = 10$ give the same polynomial $\lambda^2 - 7\lambda + 10$.

Step 3. Look for two numbers with product 10 and sum -7: these are -2 and -5. So $(\lambda - 2)(\lambda - 5) = 0$.

$$\boxed{\lambda = 2,\ 5}$$

Step 4. $2 + 5 = 7 = \text{tr}\, A$ ✓, $2 \cdot 5 = 10 = \det A$ ✓.

Stable? No: $|2| > 1$ and $|5| > 1$.

Problem 2

$$A = \begin{bmatrix} 3 & 5 \\ 0 & -1 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda - 3 & -5 \\ 0 & \lambda + 1 \end{bmatrix}$, so $\det = (\lambda - 3)(\lambda + 1) - (-5)(0) = (\lambda - 3)(\lambda + 1)$.

The 0 below the diagonal removes the off-diagonal product. This is why **triangular matrices have their diagonal entries as eigenvalues**.

$$\boxed{\lambda = 3,\ -1}$$

Check: $3 + (-1) = 2 = \text{tr}\, A$ ✓, $3 \cdot (-1) = -3 = \det A$ ✓. **Stable?** No: $|3| > 1$. ($|-1| = 1$ is also not strictly inside the unit circle.)

Problem 3

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{bmatrix}$, so $\det = \lambda(\lambda + 3) - (-1)(2) = \lambda^2 + 3\lambda + 2$.

Look for two numbers with product 2 and sum 3: these are 1 and 2. So $(\lambda + 1)(\lambda + 2) = 0$.

$$\boxed{\lambda = -1,\ -2}$$

Check: $-1 - 2 = -3 = \text{tr}\, A$ ✓, $(-1)(-2) = 2 = \det A$ ✓.

Note: in this **controllable canonical form**, the bottom row $[-2,\ -3]$ holds the coefficients of $\lambda^2 + 3\lambda + 2$ with the signs flipped.

Stable? No: $|-2| = 2 > 1$ (and $|-1| = 1$ is on the boundary).

Problem 4

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda - 1 & 2 \\ -2 & \lambda - 1 \end{bmatrix}$, so $\det = (\lambda - 1)^2 - (2)(-2) = (\lambda - 1)^2 + 4$.

Set it to zero: $(\lambda - 1)^2 = -4 \Rightarrow \lambda - 1 = \pm\sqrt{-4} = \pm 2i$.

(Same result with the formula: $\text{tr}\, A = 2$, $\det A = 1 + 4 = 5$, $\lambda = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm 4i}{2}$.)

$$\boxed{\lambda = 1 \pm 2i}$$

Check: $(1 + 2i) + (1 - 2i) = 2$ ✓, $(1 + 2i)(1 - 2i) = 1 - 4i^2 = 1 + 4 = 5$ ✓. Magnitude: $|\lambda| = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.24$. Complex eigenvalues mean rotation, and $|\lambda| > 1$ means the state spirals outward. **Stable?** No.

Problem 5

$$A = \begin{bmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{bmatrix}$$

$\lambda I - A = \begin{bmatrix} \lambda - 0.5 & -0.2 \\ -0.1 & \lambda - 0.4 \end{bmatrix}$, so $\det = (\lambda - 0.5)(\lambda - 0.4) - (0.2)(0.1) = \lambda^2 - 0.9\lambda + 0.20 - 0.02 = \lambda^2 - 0.9\lambda + 0.18$.

Quadratic formula: discriminant $= 0.9^2 - 4(0.18) = 0.81 - 0.72 = 0.09$, and $\sqrt{0.09} = 0.3$:

$$\lambda = \frac{0.9 \pm 0.3}{2} \quad \Rightarrow \quad \boxed{\lambda = 0.6,\ 0.3}$$

Check: $0.6 + 0.3 = 0.9$ ✓, $0.6 \cdot 0.3 = 0.18$ ✓. **Stable? Yes**, since $|0.6| < 1$ and $|0.3| < 1$. This is the only stable matrix among the eight problems.

Problem 6

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & -2 & 0 \\ 7 & 5 & 0.5 \end{bmatrix}$$

Step 1. Put λ on the diagonal and flip every sign. The diagonal entries become $\lambda - 1$, $\lambda - (-2) = \lambda + 2$ and $\lambda - 0.5$:

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & 0 & 0 \\ -4 & \lambda + 2 & 0 \\ -7 & -5 & \lambda - 0.5 \end{bmatrix}.$$

Step 2. Expand along row 1, $(\lambda - 1,\ 0,\ 0)$. The two zeros remove their terms:

$$\det(\lambda I - A) = +(\lambda - 1)\det \begin{bmatrix} \lambda + 2 & 0 \\ -5 & \lambda - 0.5 \end{bmatrix} - 0 \cdot (\dots) + 0 \cdot (\dots).$$

The 2×2 determinant is $(\lambda + 2)(\lambda - 0.5) - (0)(-5) = (\lambda + 2)(\lambda - 0.5)$. Therefore

$$\det(\lambda I - A) = (\lambda - 1)(\lambda + 2)(\lambda - 0.5).$$

Step 3. Set each factor to zero:

$$\boxed{\lambda = 1,\ -2,\ 0.5}$$

Step 4. Sum: $1 - 2 + 0.5 = -0.5 = \operatorname{tr} A$ ✓. Product: $1 \cdot (-2) \cdot 0.5 = -1 = \det A$ (the product of the diagonal for a triangular matrix) ✓.

Why the shortcut works: the entries below the diagonal $(4, 7, 5)$ never contributed, because each time they would appear they are multiplied by a 0 from above the diagonal.

Stable? No. $|-2| = 2 > 1$: that mode doubles in size and flips sign every step. $\lambda = 1$ never shrinks either. Only the $\lambda = 0.5$ mode dies out.

Problem 7

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & 0 & 0 \\ 0 & \lambda - 1 & -4 \\ 0 & -2 & \lambda - 3 \end{bmatrix}.$$

Expand along row 1:

$$\det = (\lambda-2)\big[(\lambda-1)(\lambda-3)-(-4)(-2)\big] = (\lambda-2)\big[\lambda^2-4\lambda+3-8\big] = (\lambda-2)(\lambda^2-4\lambda-5).$$

Factor: two numbers with product -5 and sum -4 are -5 and $+1$, so $\lambda^2-4\lambda-5 = (\lambda-5)(\lambda+1)$.

$$\boxed{\lambda = 2,\ 5,\ -1}$$

Check: $2+5-1 = 6 = 2+1+3$ ✓. $2\cdot5\cdot(-1) = -10$, and $\det A = 2\,(1\cdot3-4\cdot2) = 2(-5) = -10$ ✓. **Stable?** No.

Problem 8

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Step 1.

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & 0 & -1 \\ 0 & \lambda - 3 & 0 \\ -1 & 0 & \lambda - 2 \end{bmatrix}.$$

Step 2: choose the row. The middle row $(0,\ \lambda-3,\ 0)$ has two zeros, so only one term survives.

Step 3: sign. By the checkerboard $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$, position $(2,2)$ has sign $+$.

Step 4: expand. Delete row 2 and column 2. What remains is $\begin{bmatrix} \lambda-2 & -1 \\ -1 & \lambda-2 \end{bmatrix}$:

$$\det(\lambda I - A) = +(\lambda - 3)\det\begin{bmatrix} \lambda-2 & -1 \\ -1 & \lambda-2 \end{bmatrix}.$$

Step 5: the 2×2 determinant. $(\lambda - 2)(\lambda - 2) - (-1)(-1) = (\lambda - 2)^2 - 1$. Careful: $(-1)(-1) = +1$, and it is *subtracted*.

Step 6: factor. Use the difference of squares $a^2 - b^2 = (a - b)(a + b)$ with $a = \lambda - 2$, $b = 1$:

$$(\lambda - 2)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

(Or expand: $\lambda^2 - 4\lambda + 4 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$.)

$$\det(\lambda I - A) = (\lambda - 3)^2(\lambda - 1) \quad \Rightarrow \quad \boxed{\lambda = 1,\ 3,\ 3}$$

Step 7: check. $1 + 3 + 3 = 7 = 2 + 3 + 2$ ✓. $1 \cdot 3 \cdot 3 = 9$, and $\det A = 3\,(2 \cdot 2 - 1 \cdot 1) = 9$ (expanding along the middle row) ✓.

Same result expanding along row 1 (signs $+, -, +$):

$$\begin{aligned}\det &= +(\lambda - 2)\det\begin{bmatrix}\lambda - 3 & 0\\ 0 & \lambda - 2\end{bmatrix} - 0 + (-1)\det\begin{bmatrix}0 & \lambda - 3\\ -1 & 0\end{bmatrix}\\ &= (\lambda - 2)(\lambda - 3)(\lambda - 2) + (-1)\big[0 \cdot 0 - (\lambda - 3)(-1)\big]\\ &= (\lambda - 3)(\lambda - 2)^2 - (\lambda - 3) = (\lambda - 3)\big[(\lambda - 2)^2 - 1\big].\end{aligned}$$

It is the same expression, but with more work. That is why choosing the row with the most zeros pays off.

Stable? No: $|3| > 1$, so that mode triples every step.

Summary of Part A

#	Eigenvalues	Stable?
1	$2,\ 5$	no
2	$3,\ -1$	no
3	$-1,\ -2$	no
4	$1 \pm 2i$ ($\lvert\lambda\rvert = \sqrt{5}$)	no
5	$0.6,\ 0.3$	**yes**
6	$1,\ -2,\ 0.5$	no
7	$2,\ 5,\ -1$	no
8	$1,\ 3,\ 3$	no

Part B — Task 0: Controllable and observable subspaces

System:

$$x_{k+1} = \underbrace{\begin{bmatrix}1 & 0 & 0\\ 0 & 2 & 2\\ 0 & 1 & 3\end{bmatrix}}_{A} x_k + \underbrace{\begin{bmatrix}1\\ 0\\ 1\end{bmatrix}}_{B} u_k, \qquad y_k = \underbrace{\begin{bmatrix}1 & 1 & 1\end{bmatrix}}_{C} x_k.$$

Here $n = 3$ states, $m = 1$ input and $p = 1$ output.

B.0 Preparation: eigenvalues and eigenvectors of A

From Concepts §2.7: $\det(\lambda I - A) = (\lambda - 1)^2(\lambda - 4)$, so $\lambda = 1, 1, 4$.

Eigenvector for $\lambda = 4$: solve $(4I - A)v = 0$:

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -1 & 1 \end{bmatrix} v = 0 \;\Rightarrow\; 3v_1 = 0,\; 2v_2 - 2v_3 = 0 \;\Rightarrow\; v = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Check: $A[0,1,1]^\top = [0,\, 2+2,\, 1+3]^\top = [0,4,4]^\top = 4 \cdot [0,1,1]^\top$ ✓.

Eigenvectors for $\lambda = 1$: solve $(I - A)v = 0$:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix} v = 0 \;\Rightarrow\; -v_2 - 2v_3 = 0 \;\Rightarrow\; v_2 = -2v_3,\; v_1 \text{ free.}$$

This gives two independent eigenvectors, $[1,0,0]^\top$ and $[0,-2,1]^\top$, so the eigenspace is a plane. In particular, $[1,-2,1]^\top = [1,0,0]^\top + [0,-2,1]^\top$ is also an eigenvector for $\lambda = 1$. It will reappear in B.4.

B.1 Task 0.1 — Is the system reachable? Is it controllable?

Step 1: build $W_c = [\, B,\; AB,\; A^2B \,]$ (Concepts §5.1, $n = 3$, so we need up to A^2B).

$$B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 + 0 \\ 0 + 2 \cdot 0 + 2 \cdot 1 \\ 0 + 1 \cdot 0 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

$$A^2B = A(AB) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \cdot 2 + 2 \cdot 3 \\ 1 \cdot 2 + 3 \cdot 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \\ 11 \end{bmatrix}.$$

$$W_c = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 1 & 3 & 11 \end{bmatrix}.$$

Step 2: rank by row reduction.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 1 & 3 & 11 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 0 & 2 & 10 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(W_c) = 2 < 3 = n.$$

Cross-check with the determinant: $\det W_c = 1(2 \cdot 11 - 10 \cdot 3) - 1(0 \cdot 11 - 10 \cdot 1) + 1(0 \cdot 3 - 2 \cdot 1) = -8 + 10 - 2 = 0$ ✓.

The dependence is explicit: $A^2B = 5\,AB - 4\,B$, since $5[1,2,3] - 4[1,0,1] = [1,10,11]$. This is Cayley–Hamilton at work. With $p(\lambda) = \lambda^3 - 6\lambda^2 + 9\lambda - 4$, the minimal relation for this B happens to come from the factor $(\lambda - 1)(\lambda - 4) = \lambda^2 - 5\lambda + 4$.

Conclusion: the system is **not reachable** (not fully state controllable).

Step 3: PBH test — which eigenvalue is the problem? (Concepts §5.6)

For $\lambda = 4$:

$$[4I - A \quad B] = \begin{bmatrix} 3 & 0 & 0 & 1 \\ 0 & 2 & -2 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix}.$$

Row 1 is the only row with a nonzero first entry, so it is independent of the others. Rows 2 and 3 are not multiples of each other: $R_3 = -\frac{1}{2}R_2$ would need the last entry to be 0, but it is 1. So the rank is 3, and $\lambda = 4$ **is controllable**.

For $\lambda = 1$:

$$[I - A \quad B] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & -1 & -2 & 0 \\ 0 & -1 & -2 & 1 \end{bmatrix}.$$

$R_3 = R_1 + R_2$, so the rank is $2 < 3$. $\lambda = 1$ **is uncontrollable.**

Step 4: controllable in the sense of Def. 1.2? Controllability (steering every x_0 to the origin) would require every uncontrollable mode to vanish by itself in finite time, which needs eigenvalue 0. Here the uncontrollable eigenvalue is $\lambda = 1$. That mode neither grows nor decays, and the input cannot influence it.

> **Conclusion:** the system is **not controllable**. It is not even **stabilizable**: the uncontrollable eigenvalue 1 is not strictly inside the unit circle, so no K can make $A - BK$ asymptotically stable.

Step 5: intuition — a conserved quantity (Concepts §5.5). Take $w = [-1, -1, 1]^\top$. Then $w^\top A = -R_1 - R_2 + R_3 = [-1,\, -2 + 1,\, -2 + 3] = [-1, -1, 1] = 1 \cdot w^\top$, and $w^\top B = -1 - 0 + 1 = 0$. So $z_k := w^\top x_k = x_{3,k} - x_{1,k} - x_{2,k}$ satisfies

$$z_{k+1} = w^\top A x_k + w^\top B u_k = z_k + 0.$$

The quantity $x_3 - x_1 - x_2$ never changes, whatever input is applied. A system starting with $z_0 \neq 0$ can never reach the origin, where $z = 0$. That is exactly why it is not controllable.

Step 6: the same conclusion by direct calculation (the official solution's route). Try to steer a given x_0 to the origin.

- *How many steps can help?* Since $A^2B = 5AB - 4B$, we get $A^3B = 5A^2B - 4AB = 5(5AB - 4B) - 4AB = 21AB - 20B$, and by induction every A^kB lies in span$\{B, AB\}$. Longer input sequences add no new directions, so it is enough to look at 1 and 2 steps.
- *Two steps:* $x_2 = A^2x_0 + ABu_0 + Bu_1 = 0$ requires

$$[B \quad AB] \begin{bmatrix} u_1 \\ u_0 \end{bmatrix} = -A^2x_0, \qquad A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 6 & 10 \\ 0 & 5 & 11 \end{bmatrix}.$$

The left side always lies in the plane span$\{B, AB\}$. Because $\det A = 4 \neq 0$, A^2 is invertible, so $-A^2x_0$ can be *any* vector of $\mathbb{R}^3$ as x_0 varies. Pick an x_0 for which it lies off the plane; then the augmented matrix $[\,B \;\; AB \;\; -A^2x_0\,]$ has rank $3 > 2$, and there is no solution. *Concrete example:* $x_0 = [0, 0, 1]^\top$ gives $-A^2x_0 = [0, -10, -11]^\top$. Test with the plane equation

$x_3 = x_1 + x_2$: $-11 \neq 0 + (-10)$, so it is not in the plane. This x_0 cannot be steered to the origin.

- *One step:* $x_1 = Ax_0 + Bu_0 = 0$ needs $-Ax_0 = [u_0, 0, u_0]^\top$, which holds only for special x_0.

Hence there are initial states that can never be brought to the origin: **not controllable**, in agreement with Steps 3–5. (Check with Step 5: $x_0 = [0, 0, 1]$ has $z_0 = 1 - 0 - 0 = 1 \neq 0$.)

Python check (`python/task0_reachability_observability.py`):

```
Eigenvalues of A: [4. 1. 1.]
Reachability matrix R = [B, AB, A^2 B]:
 [[ 1.  1.  1.]
 [ 0.  2. 10.]
 [ 1.  3. 11.]]
rank(R) = 2  (n = 3) -> NOT reachable
PBH test for each distinct eigenvalue:
  lambda = 1: rank[lambda I - A, B] = 2 -> uncontrollable mode
  lambda = 4: rank[lambda I - A, B] = 3 -> controllable mode
```

B.2 Task 0.2 — Reachable subspace and target states

Step 1: reachable subspace $=$ im(W_c) (Concepts §5.4). Since A^2B depends on the first two columns,

$$\mathcal{R} = \text{span}\left\{ \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\2\\3\end{bmatrix} \right\},$$

which is a plane through the origin (dimension 2 = rank).

Step 2: describe the plane with one equation using the cross product (Concepts §1.6):

$$\begin{bmatrix}1\\0\\1\end{bmatrix} \times \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}0\cdot 3 - 1\cdot 2\\1\cdot 1 - 1\cdot 3\\1\cdot 2 - 0\cdot 1\end{bmatrix} = \begin{bmatrix}-2\\-2\\2\end{bmatrix} \propto \begin{bmatrix}-1\\-1\\1\end{bmatrix}.$$

So the plane is $-x_1 - x_2 + x_3 = 0$:

$$\boxed{\mathcal{R} = \{\, x \in \mathbb{R}^3 : x_3 = x_1 + x_2 \,\}}$$

Check: B: $1 = 1 + 0$ ✓; AB: $3 = 1 + 2$ ✓; A^2B: $11 = 1 + 10$ ✓. This normal vector is exactly the conserved-quantity vector w from B.1 Step 5. Starting from $x_0 = 0$ we have $z = 0$ forever, so we stay in the plane $x_3 - x_1 - x_2 = 0$.

Step 3: target $x_f^{(1)} = [4, 2, 3]^\top$. $x_1 + x_2 = 4 + 2 = 6 \neq 3 = x_3$. The target is not in $\mathcal{R}$.

$x_f^{(1)}$ **is not reachable** from the origin, with any number of steps or any inputs.

(Rank test: rank$[W_c \;\; x_f^{(1)}] = 3 \neq 2$.)

Step 4: target $x_f^{(2)} = [6, 2, 8]^\top$. $x_1 + x_2 = 6 + 2 = 8 = x_3$ ✓. **Reachable.**

Step 5: minimum number of steps. From $x_0 = 0$, after N steps $x_N = \sum_{j=0}^{N-1} A^{N-1-j} B u_j$.

Try $N = 1$*:* $x_1 = Bu_0 = [u_0,\ 0,\ u_0]^\top$. The second component is always 0, but we need 2. **Impossible.**

Try $N = 2$*:* $x_2 = AB\,u_0 + B\,u_1$:

$$u_0 \begin{bmatrix}1\\2\\3\end{bmatrix} + u_1 \begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}6\\2\\8\end{bmatrix} \quad\Longleftrightarrow\quad \begin{cases} u_0 + u_1 = 6 \\ 2u_0 = 2 \\ 3u_0 + u_1 = 8 \end{cases}$$

- Row 2: $u_0 = 1$.
- Row 1: $u_1 = 6 - 1 = 5$.
- Row 3 (consistency check): $3 \cdot 1 + 5 = 8$ ✓.

  $x_f^{(2)}$ **is reachable in a minimum of** $N = 2$ **steps, with** $u_0 = 1,\ u_1 = 5$**.** (The solution is unique, because B and AB are independent.)

Step 6: verify by simulation.

$$x_1 = A \cdot 0 + B \cdot 1 = \begin{bmatrix}1\\0\\1\end{bmatrix}, \qquad x_2 = Ax_1 + B \cdot 5 = \begin{bmatrix}1\\2\\3\end{bmatrix} + \begin{bmatrix}5\\0\\5\end{bmatrix} = \begin{bmatrix}6\\2\\8\end{bmatrix} \checkmark$$

Python check:

```
x_f1 = [4. 2. 3.]: reachable? False
x_f2 = [6. 2. 8.]: reachable? True
  minimum number of steps N = 2, inputs u = [1. 5.]
    x_1 = [1. 0. 1.]
    x_2 = [6. 2. 8.]
```

B.3 Task 0.3 — Is the system observable?

Step 1: build $W_o = [\,C;\ CA;\ CA^2\,]$ (Concepts §6.1). A row vector times A is a weighted sum of the rows of A.

$$C = [1 \quad 1 \quad 1]\,.$$

$$CA = 1 \cdot [1,0,0] + 1 \cdot [0,2,2] + 1 \cdot [0,1,3] = [1 \quad 3 \quad 5]\,.$$

$$CA^2 = (CA)A = 1 \cdot [1,0,0] + 3 \cdot [0,2,2] + 5 \cdot [0,1,3] = [1,\ 6+5,\ 6+15] = [1 \quad 11 \quad 21]\,.$$

$$W_o = \begin{bmatrix}1 & 1 & 1\\ 1 & 3 & 5\\ 1 & 11 & 21\end{bmatrix}.$$

Step 2: rank.

$$\xrightarrow{R_2-R_1,\ R_3-R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 10 & 20 \end{bmatrix} \xrightarrow{R_3-5R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(W_o) = 2 < 3.$$

(Again $CA^2 = 5\,CA - 4\,C$: $5[1,3,5] - 4[1,1,1] = [1,11,21]$ ✓.)

Conclusion: the system is **not observable**.

Step 3: PBH test.

$\lambda = 4$: $\begin{bmatrix} 4I - A \\ C \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Rows 1, 2 and 4 give det $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & -2 \\ 1 & 1 & 1 \end{bmatrix} = 3(2\cdot 1 - (-2)\cdot 1) = 12 \neq 0$, so the rank is 3. **Observable mode.**

$\lambda = 1$: $\begin{bmatrix} I - A \\ C \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix}$. Rows 2 and 3 are equal and row 1 is zero, so only $[0,-1,-2]$ and $[1,1,1]$ are independent. The rank is $2 < 3$. **Unobservable mode.**

Equivalently, the $\lambda = 1$ eigenvector $v = [1,-2,1]^\top$ has $Cv = 1 - 2 + 1 = 0$: the sensor cannot see it. The system is also **not detectable**, because the unobservable eigenvalue 1 is not inside the unit circle.

Python check:

```
Observability matrix O = [C; CA; CA^2]:
 [[ 1.  1.  1.]
 [ 1.  3.  5.]
 [ 1. 11. 21.]]
rank(O) = 2  (n = 3) -> NOT observable
  lambda = 1: rank[lambda I - A; C] = 2 -> unobservable mode
  lambda = 4: rank[lambda I - A; C] = 3 -> observable mode
```

B.4 Task 0.4 — Observable subspace and initial states

Step 1: unobservable subspace $= \ker(W_o)$**.** Solve $W_o v = 0$. Use the two independent rows:

$$v_1 + v_2 + v_3 = 0 \qquad (1)$$
$$v_1 + 3v_2 + 5v_3 = 0 \qquad (2)$$

$(2) - (1)$: $2v_2 + 4v_3 = 0 \Rightarrow v_2 = -2v_3$. Into (1): $v_1 = -v_2 - v_3 = 2v_3 - v_3 = v_3$. With $v_3 = t$: $v = t\,[1,-2,1]^\top$.

Check with row 3: $1 - 22 + 21 = 0$ ✓.

$$\boxed{\ker(W_o) = \text{span}\{[1,\,-2,\,1]^\top\}} \quad \text{(a line, dimension } 3 - 2 = 1)$$

Step 2: observable subspace $= \ker(W_o)^\perp$ = row space of W_o:

$$\boxed{\mathcal{O} = \{\, x : x_1 - 2x_2 + x_3 = 0 \,\} = \text{span}\{[1,1,1]^\top,\ [1,3,5]^\top\}}$$

Check: $[1,1,1]$: $1-2+1=0$ ✓; $[1,3,5]$: $1-6+5=0$ ✓.

Step 3: initial state $x_0^{(1)} = [1,2,3]^\top$.

- Test for membership in $\mathcal{O}$: $1 - 2\cdot 2 + 3 = 0$ ✓. So $x_0^{(1)}$ lies in the observable subspace, and its unobservable part is zero.
- Its outputs: $\vec{y} = W_o x_0^{(1)} = [1+2+3,\ 1+6+15,\ 1+22+63]^\top = [6,\ 22,\ 86]^\top$, which is nonzero.

  $x_0^{(1)}$ **can be observed**: it is the unique state in the observable subspace that produces the outputs $6, 22, 86$. *Fine print:* since the system is not observable, any $x_0^{(1)} + c\,[1,-2,1]^\top$ would produce the same outputs. The measurements identify the observable component, and $x_0^{(1)}$ consists only of that component.

Explicit decomposition (as in the official solution): write $x_0^{(1)} = m[1,1,1]^\top + n[1,3,5]^\top$. The first two rows give $m+n=1$ and $m+3n=2$, so $n=\frac{1}{2}$ and $m=\frac{1}{2}$. Check row 3: $\frac{1}{2}+\frac{5}{2}=3$ ✓. So

$$x_0^{(1)} = \tfrac{1}{2}\begin{bmatrix}1\\1\\1\end{bmatrix} + \tfrac{1}{2}\begin{bmatrix}1\\3\\5\end{bmatrix} \in \mathcal{O}.$$

Step 4: initial state $x_0^{(2)} = [-2,4,-2]^\top$.

- $x_0^{(2)} = -2\,[1,-2,1]^\top$, so it lies entirely in $\ker(W_o)$.
- Outputs: $y_0 = -2+4-2 = 0$; $y_1 = -2+12-10 = 0$; $y_2 = -2+44-42 = 0$.
- Because $[1,-2,1]$ is an eigenvector for $\lambda = 1$, the state never moves ($x_k = x_0^{(2)}$ for all k), and $Cx_k = 0$ forever.

  $x_0^{(2)}$ **cannot be observed**: it produces $y_k = 0$ for all k, exactly like $x_0 = 0$.

Python check:

```
Basis of UNobservable subspace (kernel of O): [-0.4082  0.8165 -0.4082]   (= multiple of [1,
x0_1 = [1. 2. 3.]
  output sequence y_0..y_2 = [ 6. 22. 86.]
  -> lies in the observable subspace: can be observed
x0_2 = [-2.  4. -2.]
  output sequence y_0..y_2 = [0. 0. 0.]
  -> lies entirely in the unobservable subspace: cannot be observed
```

Task 0 — summary

Question	Answer
Eigenvalues of A	$1, 1, 4$
rank W_c	2 — **not reachable**
Uncontrollable mode (PBH)	$\lambda = 1$ — **not controllable**, not stabilizable
Reachable subspace	plane $x_3 = x_1 + x_2$
$x_f^{(1)} = [4,2,3]$	not reachable

Question	Answer
$x_f^{(2)} = [6, 2, 8]$	reachable, min. $N = 2$, $(u_0, u_1) = (1, 5)$
rank W_o	2 — **not observable**
Unobservable mode (PBH)	$\lambda = 1$ — not detectable
Unobservable subspace	span$\{[1, -2, 1]\}$
Observable subspace	plane $x_1 - 2x_2 + x_3 = 0$
$x_0^{(1)} = [1, 2, 3]$	observable ($y = 6, 22, 86$)
$x_0^{(2)} = [-2, 4, -2]$	not observable ($y \equiv 0$)

Part C — Task 1: Tracking error for a state-feedback controller

Setting: $x_{k+1} = Ax_k + Bu_k$, controller $u_k = -K(x_k - x_{\text{ref}})$ with constant x_{ref}, and K such that $A - BK$ is asymptotically stable.

C.0 Setting up the closed loop

Substitute the controller into the system:

$$\begin{aligned} x_{k+1} &= Ax_k + B(-K(x_k - x_{\text{ref}})) \\ &= Ax_k - BKx_k + BKx_{\text{ref}} \\ &= \underbrace{(A - BK)}_{=:M} x_k + BKx_{\text{ref}}. \end{aligned}$$

So the closed loop is $x_{k+1} = Mx_k + c$ with constant $c = BKx_{\text{ref}}$, and by assumption all eigenvalues of M satisfy $|\lambda| < 1$.

C.1 Task 1.1 — $I - (A - BK)$ is invertible and $x_k \to x_\infty$

Part (a): invertibility of $I - M$ Proof by contradiction.

1. Suppose $I - M$ is **not** invertible.
2. Then there is a vector $v \neq 0$ with $(I - M)v = 0$ (Concepts §1.7).
3. Expand: $v - Mv = 0$, i.e. $Mv = v = 1 \cdot v$.
4. So v is an eigenvector of M with eigenvalue $\lambda = 1$.
5. But M is asymptotically stable, so every eigenvalue satisfies $|\lambda| < 1$. Since $|1| = 1$, this is a contradiction.
6. Therefore $I - M = I - (A - BK)$ **is invertible**. ■

   Remember: $I - M$ is singular $\iff$ 1 is an eigenvalue of M.

Part (b): convergence to $x_\infty = (I - (A - BK))^{-1}BKx_{\text{ref}}$ Step 1: find the equilibrium. A limit must be an equilibrium (Concepts §4.3), so set $x_{k+1} = x_k = x_\infty$:

$$x_\infty = Mx_\infty + BKx_{\text{ref}} \implies x_\infty - Mx_\infty = BKx_{\text{ref}} \implies (I - M)\, x_\infty = BKx_{\text{ref}}.$$

By part (a), $I - M$ is invertible, so there is exactly one equilibrium:

$$x_\infty = (I - M)^{-1} BK x_{\text{ref}} = \big(I - (A - BK)\big)^{-1} BK x_{\text{ref}}.$$

Step 2: show that the trajectory actually converges to it (shift trick, Concepts §4.4). Define $d_k := x_k - x_\infty$:

$$\begin{aligned} d_{k+1} &= x_{k+1} - x_\infty \\ &= (Mx_k + BKx_{\text{ref}}) - (Mx_\infty + BKx_{\text{ref}}) \quad \text{(both satisfy the same update)} \\ &= M(x_k - x_\infty) \\ &= Md_k. \end{aligned}$$

By induction, $d_k = M^k d_0$. Since M is asymptotically stable, $M^k \to 0$, hence $d_k \to 0$ for **every** x_0:

$$\lim_{k\to\infty} x_k = x_\infty = \big(I - (A - BK)\big)^{-1} BK x_{\text{ref}}. \qquad \blacksquare$$

Alternative proof (via the lecture's Proposition 1.1) The closed loop is a system with matrix M and constant "input" BKx_{ref}. By the solution formula,

$$x_k = M^k x_0 + \sum_{j=0}^{k-1} M^{k-1-j} BK x_{\text{ref}} = M^k x_0 + \Big(\sum_{i=0}^{k-1} M^i\Big) BK x_{\text{ref}}.$$

As $k \to \infty$: $M^k x_0 \to 0$, and by the geometric series (Concepts §4.2) $\sum_{i=0}^{\infty} M^i = (I - M)^{-1}$. This gives the same x_∞.

C.2 Task 1.2 — Asymptotic tracking error and when it can be zero

Step 1: the one-dimensional case (as the hint suggests) Let $A = a$, $B = b$, $K = k$, all scalars. The closed loop is $x_{k+1} = (a - bk)x_k + bk\, x_{\text{ref}}$.

- **Stability:** $|a - bk| < 1$.
- **Limit:** from C.1, $x_\infty = \dfrac{bk}{1 - (a - bk)}\, x_{\text{ref}} = \dfrac{bk}{1 - a + bk}\, x_{\text{ref}}$.
- **Error:** put both terms over a common denominator:

$$\begin{aligned} x_\infty - x_{\text{ref}} &= \frac{bk}{1 - a + bk} x_{\text{ref}} - \frac{1 - a + bk}{1 - a + bk} x_{\text{ref}} \\ &= \frac{bk - 1 + a - bk}{1 - a + bk}\, x_{\text{ref}} \\ &= \boxed{\frac{a - 1}{1 - a + bk}\, x_{\text{ref}}} \end{aligned}$$

Observation: the numerator $a - 1$ **does not contain** k. Therefore:

- If $a = 1$: the error is 0 for every stabilizing k ✓.
- If $a \neq 1$ (and $x_{\text{ref}} \neq 0$): the error is **never** zero, whatever k is.

Can the error at least be made small? A larger bk makes the denominator larger, but stability restricts it: $|a-bk|<1 \iff a-1<bk<a+1$. So $1-a+bk<2$, and the error is bounded below in size by $\frac{|a-1|}{2}|x_{\text{ref}}|$. It cannot be made arbitrarily small.

Numerical example: $a=0.5$, $b=1$. Stability requires $bk \in (-0.5,\ 1.5)$.

k	$a-bk$ (closed-loop pole)	$x_\infty/x_{\text{ref}} = \frac{bk}{1-a+bk}$	error $/x_{\text{ref}}$
0.5	0	$0.5/1.0 = 0.5$	-0.5
1.0	-0.5	$1.0/1.5 \approx 0.667$	≈ -0.333
1.4	-0.9	$1.4/1.9 \approx 0.737$	≈ -0.263
$\to 1.5$	$\to -1$ (stability limit)	$\to 0.75$	$\to -0.25$

You never get closer than 75 % of the way to the target.

Remark on the official solution. It notes that the error "decays to zero as $k \to \infty$". That is true for the formula $\frac{a-1}{1-a+bk}$ on its own, i.e. if the gain could grow without limit (as in continuous time). In discrete time, however, the closed loop is only stable for $bk < a+1$, so the gain cannot grow beyond that, and the bound above applies. Both statements agree that a larger gain means a smaller error; the discrete-time stability limit just caps how far you can go.

Step 2: the general (matrix) case Use the trick $x_{\text{ref}} = (I-M)^{-1}(I-M)\,x_{\text{ref}}$, which is valid because $I-M$ is invertible:

$$\begin{aligned} x_\infty - x_{\text{ref}} &= (I-M)^{-1}BKx_{\text{ref}} - (I-M)^{-1}(I-M)x_{\text{ref}} \\ &= (I-M)^{-1}\big[BK-(I-M)\big]x_{\text{ref}} \\ &= (I-M)^{-1}\big[BK-I+A-BK\big]x_{\text{ref}} \qquad (M=A-BK) \\ &= (I-M)^{-1}(A-I)\,x_{\text{ref}}. \end{aligned}$$

$$\boxed{x_\infty - x_{\text{ref}} = \big(I-A+BK\big)^{-1}(A-I)\,x_{\text{ref}}}$$

For $n=1$ this reduces to the scalar formula above. The official solution writes the same result as $x_\infty - x_{\text{ref}} = -\big(I-(A-BK)\big)^{-1}(I-A)x_{\text{ref}}$; the two minus signs cancel.

Step 3: when is the error zero? $(I-A+BK)^{-1}$ is invertible, so it maps only the zero vector to zero. Therefore

$$x_\infty - x_{\text{ref}} = 0 \iff (A-I)\,x_{\text{ref}} = 0 \iff A\,x_{\text{ref}} = x_{\text{ref}}.$$

Answer to Task 1.2. The asymptotic tracking error is $x_\infty - x_{\text{ref}} = (I-A+BK)^{-1}(A-I)x_{\text{ref}}$. It is zero **if and only if** $Ax_{\text{ref}} = x_{\text{ref}}$, i.e. x_{ref} is already an equilibrium of the *uncontrolled* system ($u=0$), or equivalently $x_{\text{ref}} \in \ker(A-I)$ (an eigenvector of A for eigenvalue 1, or $x_{\text{ref}}=0$). - If this holds, **any** stabilizing K achieves $x_\infty = x_{\text{ref}}$. - If it does not hold, **no** choice of K achieves zero error. The condition does not depend on K at all. - Zero error for **every** reference requires $A=I$ (in 1-D: $a=1$).

Step 4: interpretation (why this happens) To hold the state at x_{ref}, the input must satisfy $x_{\text{ref}} = Ax_{\text{ref}} + B\bar{u}$, i.e. $B\bar{u} = (I - A)x_{\text{ref}}$. This input is nonzero unless $Ax_{\text{ref}} = x_{\text{ref}}$. But at $x = x_{\text{ref}}$ the P-controller outputs $u = -K \cdot 0 = 0$. The controller can only produce input by keeping an error, so the system settles where the error is just large enough to supply the needed input. This is the classic **steady-state error of a P-controller** (Concepts §7.3).

This motivates Task 2 (rescale the reference with a prefilter F) and Task 3 (add an integrator, i.e. a PI controller).

Step 5: numerical verification (`python/task1_tracking_error.py`)

Case	Simulated x_∞	Formula
1-D: $a = 0.5,\ b = 1,\ k = 0.5,\ x_{\text{ref}} = 1$	0.5	$\frac{0.5}{1.0} = 0.5$ ✓
System a) of the numerical experiment, $K = \begin{bmatrix} 0 & 0.2 \\ 0.6 & 0 \end{bmatrix}$, $x_{\text{ref}} = [1, 0]^\top$	$[0.931,\ 0.103]$	$[0.931,\ 0.103]$ ✓
Error for the same case	$[-0.069,\ 0.103]$	$(I-A+BK)^{-1}(A-I)x_{\text{ref}} = [-0.069,\ 0.103]$ ✓

For system a), the closed-loop eigenvalues are 0.445 and -0.045 (stable), but $A[1, 0]^\top = [0.9, 0.1]^\top \neq [1, 0]^\top$, so a steady-state error remains, as the theory predicts.

Part D — Task 2: State feedback + reference prefilter

Setting: $x_{k+1} = Ax_k + Bu_k$, controller $u_k = -K(x_k - Fx_{\text{ref}})$ with constant x_{ref}, a prefilter matrix F, and K such that $A - BK$ is asymptotically stable. Theory: Concepts §7.4–7.5.

D.0 Setting up the closed loop

Substitute the controller into the system:

$$\begin{aligned} x_{k+1} &= Ax_k + B(-K(x_k - Fx_{\text{ref}})) \\ &= Ax_k - BKx_k + BKFx_{\text{ref}} \\ &= \underbrace{(A - BK)}_{=:M} x_k + BKF\, x_{\text{ref}}. \end{aligned}$$

The only difference from Task 1 (Part C.0) is the constant term: $BKFx_{\text{ref}}$ instead of BKx_{ref}.

D.1 Task 2.1 — The limit $x_\infty = (I - (A - BK))^{-1}BKFx_{\text{ref}}$

Step 1: rename the target. Define the modified reference

$$\tilde{x}_{\text{ref}} := F\, x_{\text{ref}}.$$

F and x_{ref} are constant, so $\tilde{x}_{\text{ref}}$ is a constant vector too. The controller becomes

$$u_k = -K(x_k - \tilde{x}_{\text{ref}}),$$

which is **exactly the Task 1 controller**, only with $\tilde{x}_{\text{ref}}$ as its reference.

Step 2: apply Task 1.1. All assumptions of Task 1.1 hold: same A, B, K, $A-BK$ asymptotically stable, constant reference. So, by Part C.1:

- $I - (A - BK)$ is invertible (because 1 is not an eigenvalue of the stable matrix $A - BK$), and
- $x_k \to (I - (A - BK))^{-1} BK\, \tilde{x}_{\text{ref}}$ for every x_0.

Step 3: substitute back $\tilde{x}_{\text{ref}} = F x_{\text{ref}}$:

$$\boxed{x_\infty = \big(I - (A - BK)\big)^{-1} BKF\, x_{\text{ref}}} \qquad \blacksquare$$

Directly, without Task 1: the equilibrium of $x_{k+1} = Mx_k + BKFx_{\text{ref}}$ solves $(I - M)x_\infty = BKFx_{\text{ref}}$, and the deviation $d_k = x_k - x_\infty$ satisfies $d_{k+1} = Md_k \to 0$. This is the same shift trick as in C.1.

D.2 Task 2.2 — Tracking error and when it can be zero

Step 1: the error formula As in Task 1, write $x_{\text{ref}} = (I - M)^{-1}(I - M)x_{\text{ref}}$:

$$\begin{aligned} x_\infty - x_{\text{ref}} &= (I - M)^{-1} BKF x_{\text{ref}} - (I - M)^{-1}(I - M)x_{\text{ref}} \\ &= (I - M)^{-1}\big[BKF - I + M\big]x_{\text{ref}} \\ &= (I - M)^{-1}\big[BKF - I + A - BK\big]x_{\text{ref}} \\ &= \big(I - (A - BK)\big)^{-1}\big[(A - BK + BKF) - I\big]x_{\text{ref}}. \end{aligned}$$

Equivalently, as in the official solution: $x_\infty - x_{\text{ref}} = \big[(I - (A - BK))^{-1} BKF - I\big]x_{\text{ref}}$.

Check: with $F = I$ the bracket is $A - I$, which is the Task 1 result ✓.

Step 2: the zero-error condition $(I - M)^{-1}$ is invertible, so the error is zero exactly when the bracket kills x_{ref}:

$$x_\infty = x_{\text{ref}} \iff BKFx_{\text{ref}} = (I - A + BK)x_{\text{ref}} \iff \boxed{(A - BK + BKF)\, x_{\text{ref}} = x_{\text{ref}}}$$

The important difference from Task 1: the condition now contains F, which we are free to choose. In Task 1 the condition $Ax_{\text{ref}} = x_{\text{ref}}$ did not contain anything we could design.

Step 3: the one-dimensional case (hint) With scalars a, b, k, f, the condition reads $(a - bk + bkf)x_{\text{ref}} = x_{\text{ref}}$. For $x_{\text{ref}} \neq 0$, divide by x_{ref}:

$$a - bk + bkf = 1 \implies bkf = 1 - a + bk \implies \boxed{f = \frac{1 - a + bk}{bk} = 1 + \frac{1 - a}{bk}}$$

Requirements: $b \neq 0$ (the input must affect the state), $k \neq 0$ (otherwise there is no feedback through which f could act), and $|a - bk| < 1$ (stability). Then **zero error for every reference**, for any stabilizing k.

Interpretation: from Task 1, $x_\infty = \frac{bk}{1-a+bk} x_{\text{ref}}$, i.e. the closed loop only delivers the fraction $\frac{bk}{1-a+bk}$ of the target. The prefilter multiplies the target by the inverse of that fraction, so the controller **aims higher on purpose** and lands exactly on the target.

Example ($a = 0.5,\ b = 1,\ k = 0.5$ from Task 1): $f = \frac{1-0.5+0.5}{0.5} = 2$. Task 1 settled at $0.5\, x_{\text{ref}}$; aiming at $2x_{\text{ref}}$ gives $x_\infty = 0.5 \cdot 2x_{\text{ref}} = x_{\text{ref}}$ ✓ (`python/task2_prefilter.py`).

Step 4: the general (matrix) case **(i) For one particular reference:** zero error $\iff$ x_{ref} is an eigenvector of $A - BK + BKF$ with eigenvalue 1.

(ii) For every reference at once: we need $A - BK + BKF = I$. Solve for F step by step:

$$\begin{aligned} A - BK + BKF &= I \\ BKF - BK &= I - A \\ BK(F - I) &= I - A \\ F - I &= (BK)^{-1}(I - A) \qquad \text{(if } BK \text{ is invertible)} \\ F &= (BK)^{-1}(I - A) + I. \end{aligned}$$

$$\boxed{F = (BK)^{-1}(I - A) + I}$$

Verification: $BKF = BK(BK)^{-1}(I - A) + BK = I - A + BK$, so $A - BK + BKF = A - BK + I - A + BK = I$ ✓.

When is BK invertible? BK is $n \times n$, and $\operatorname{rank}(BK) \leq \operatorname{rank} B \leq m$. Invertibility needs rank n, so

$$m \geq n, \quad \operatorname{rank} B = n, \quad \text{and } K \text{ chosen stabilizing with } BK \text{ invertible.}$$

(iii) Fewer inputs than states ($m < n$). The general formula is impossible. Can a particular x_{ref} still be reached? The condition $BKFx_{\text{ref}} = (I - A)x_{\text{ref}} + BKx_{\text{ref}}$ has a left side in $\operatorname{im} B$, and $BKx_{\text{ref}} \in \operatorname{im} B$ as well. So we need

$$(I - A)\, x_{\text{ref}} \in \operatorname{im} B,$$

i.e. x_{ref} must be **holdable** by a constant input (Concepts §7.4). If it is, with $B\bar{u} = (I - A)x_{\text{ref}}$, then any F with $KFx_{\text{ref}} = \bar{u} + Kx_{\text{ref}}$ works. If it is not, no prefilter (and no other controller) can reach it.

Step 5: interpretation — the prefilter is a feedforward

$$u_k = -K(x_k - Fx_{\text{ref}}) = -K(x_k - x_{\text{ref}}) + \underbrace{K(F - I)x_{\text{ref}}}_{u_{\text{ff}}}.$$

With F from Step 4: $Bu_{\text{ff}} = BK(F - I)x_{\text{ref}} = (I - A)x_{\text{ref}}$. That is precisely the input needed to **hold** the state at x_{ref} (Concepts §7.4). The feedback part is zero at the target, and the feedforward part supplies the holding input.

Answer to Task 2.2. $x_\infty - x_{\text{ref}} = (I-(A-BK))^{-1}\big[(A-BK+BKF)-I\big]x_{\text{ref}}$, which is zero iff $(A - BK + BKF)x_{\text{ref}} = x_{\text{ref}}$. - **1-D:** choose $f = \dfrac{1-a+bk}{bk}$ (needs $b \neq 0$, stabilizing $k \neq 0$): zero error for every x_{ref}. - **Matrix, all references:** if rank $B = n$ ($m \geq n$) and BK is invertible: $F = (BK)^{-1}(I - A) + I$. - $m < n$**:** possible only for references with $(I - A)x_{\text{ref}} \in \text{im}\, B$.

D.3 Task 2.3 — Comparing Task 1 b) and Task 2 b)

	Task 1: $u = -K(x - x_{\text{ref}})$	Task 2: $u = -K(x - Fx_{\text{ref}})$
zero-error condition	$Ax_{\text{ref}} = x_{\text{ref}}$	$(A - BK + BKF)x_{\text{ref}} = x_{\text{ref}}$
depends on the design?	**no**, K drops out	**yes**, F can be chosen
zero error for all x_{ref}	only if $A = I$	if rank $B = n$ (choose F as above)
1-D	only if $a = 1$	always, with $f = \frac{1-a+bk}{bk}$
input at the target	0	holding input u_{ff} (feedforward)

Discussion.

1. **Separation of jobs.** In Task 1, the gain K has to do everything, and it cannot fix the steady state. In Task 2, K is chosen for **stability and speed** (the eigenvalues of $A - BK$), and F is chosen afterwards for **steady-state accuracy**. The two design goals are decoupled.
2. **No extra dynamics.** The prefilter adds no states. The transient is governed by the same $A-BK$, and only the target is shifted. In the numerical experiment (Part F) with $A-BK = 0.5I$, the state moves along a straight line, $x_k = (1 - 0.5^k)x_{\text{ref}}$, with no overshoot.
3. **It relies on the model.** $F = (BK)^{-1}(I-A)+I$ contains A and B. If the real plant differs, the feedforward is wrong, and nothing in the loop notices. *1-D example:* $f = 2$ was designed for $a = 0.5$. If the real plant has $a = 0.55$, then $x_\infty = \frac{bkf}{1-a+bk}x_{\text{ref}} = \frac{0.5\cdot 2}{1-0.55+0.5}x_{\text{ref}} = \frac{1}{0.95}x_{\text{ref}} = 1.053\, x_{\text{ref}}$, a 5 % error. *Matrix example (Part F):* with a 3 % model error, systems a) and b) get errors of about 0.06 with the prefilter, compared to 0.23–0.26 without it, and 0 with the integrator.
4. **It needs enough inputs.** The general formula needs BK invertible ($m \geq n$). For the single-input DC motor (system c), only holdable references can be reached.
5. **Constant disturbances** (e.g. an unknown load) are not compensated either, for the same reason as point 3. This motivates Task 3: an integrator *measures* the remaining error and corrects it.

Part E — Task 3: State feedback + integrator

Setting: $u_k = -K_p e_k - K_I e_k^i = -\begin{bmatrix}K_p & K_I\end{bmatrix}\begin{bmatrix}e_k \\ e_k^i\end{bmatrix}$ with $e_k := x_k - x_{\text{ref}}$ (constant x_{ref}), $e_{k+1}^i = e_k^i + T_s e_k$, $T_s > 0$, and $K_p, K_I \in \mathbb{R}^{m\times n}$. Theory: Concepts §7.6–7.7.

E.1 Task 3.1 — The augmented state-space model

We want one model with state $z_k = \begin{bmatrix}x_k \\ e_k^i\end{bmatrix} \in \mathbb{R}^{2n}$ and "input" x_{ref}.

Step 1: write the two update equations on top of each other.

$$\begin{bmatrix} x_{k+1} \\ e^i_{k+1} \end{bmatrix} = \begin{bmatrix} Ax_k + Bu_k \\ e^i_k + T_s e_k \end{bmatrix}.$$

Step 2: insert the controller $u_k = -K_p e_k - K_I e^i_k$ in the first row:

$$= \begin{bmatrix} Ax_k + B(-K_p e_k - K_I e^i_k) \\ e^i_k + T_s e_k \end{bmatrix}.$$

Step 3: insert the error $e_k = x_k - x_{\text{ref}}$ in both rows:

$$= \begin{bmatrix} Ax_k - BK_p(x_k - x_{\text{ref}}) - BK_I e^i_k \\ e^i_k + T_s(x_k - x_{\text{ref}}) \end{bmatrix}.$$

Step 4: multiply out and sort each row into (x_k terms) + (e^i_k terms) + (x_{ref} terms):

$$= \begin{bmatrix} (A - BK_p)\, x_k \;-\; BK_I\, e^i_k \;+\; BK_p\, x_{\text{ref}} \\ T_s\, x_k \;+\; e^i_k \;-\; T_s\, x_{\text{ref}} \end{bmatrix}.$$

Step 5: read off the block matrices. The coefficient of x_k in row 2 is $T_s I$, and that of e^i_k is I (identity, size n):

$$\boxed{\begin{bmatrix} x_{k+1} \\ e^i_{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} A - BK_p & -BK_I \\ T_s I & I \end{bmatrix}}_{\tilde{A} \in \mathbb{R}^{2n \times 2n}} \begin{bmatrix} x_k \\ e^i_k \end{bmatrix} + \underbrace{\begin{bmatrix} BK_p \\ -T_s I \end{bmatrix}}_{\tilde{B} \in \mathbb{R}^{2n \times n}} x_{\text{ref}}}$$

Check by block multiplication (Concepts §7.6): row 1 of $\tilde{A}z_k + \tilde{B}x_{\text{ref}}$ is $(A - BK_p)x_k - BK_I e^i_k + BK_p x_{\text{ref}}$ ✓, and row 2 is $T_s x_k + e^i_k - T_s x_{\text{ref}}$ ✓. The dynamics matrix is the one given in the hint.

Numerical check (`python/task3_integrator.py`): simulating the controller directly and simulating $z_{k+1} = \tilde{A}z_k + \tilde{B}x_{\text{ref}}$ give the same trajectory, with a maximum difference of $2 \cdot 10^{-16}$ (rounding).

E.2 Task 3.2 — Asymptotic stability ⇒ zero tracking error

Assume $\tilde{A}$ is asymptotically stable. The augmented model $z_{k+1} = \tilde{A}z_k + \tilde{B}x_{\text{ref}}$ has exactly the form of Task 1 (a stable matrix plus a constant input), so the Task 1 arguments carry over.

Step 1: $I - \tilde{A}$ **is invertible.** If not, $\tilde{A}v = v$ for some $v \neq 0$, so 1 would be an eigenvalue of $\tilde{A}$. This contradicts $|\lambda| < 1$.

Step 2: z_k **converges** (Task 1.1 / Proposition 1.1 with $M = \tilde{A}$ and constant term $\tilde{B}x_{\text{ref}}$):

$$z_k \to \bar{z} = \begin{bmatrix} x_\infty \\ e^i_\infty \end{bmatrix} = (I - \tilde{A})^{-1} \tilde{B}\, x_{\text{ref}},$$

and the limit satisfies the equilibrium equation $\bar{z} = \tilde{A}\bar{z} + \tilde{B}x_{\text{ref}}$.

Step 3: write the equilibrium equation block by block.

$$\begin{aligned} \text{row 1:} \quad & x_\infty = (A - BK_p)x_\infty - BK_I e^i_\infty + BK_p x_{\text{ref}}, \\ \text{row 2:} \quad & e^i_\infty = T_s x_\infty + e^i_\infty - T_s x_{\text{ref}}. \end{aligned}$$

Step 4: evaluate the integrator row (the hint). Subtract e^i_∞ from both sides of row 2:

$$0 = T_s x_\infty - T_s x_{\text{ref}} = T_s(x_\infty - x_{\text{ref}}).$$

Since $T_s > 0$, divide by T_s:

$$\boxed{x_\infty = x_{\text{ref}}} \qquad \blacksquare$$

Nothing about A, B, K_p or K_I was used except stability. **The zero error does not depend on knowing the model exactly** (unlike Task 2).

Step 5 (extra): what the integrator converges to. Put $x_\infty = x_{\text{ref}}$ into row 1. The K_p terms cancel:

$$x_{\text{ref}} = Ax_{\text{ref}} - BK_I e^i_\infty \quad \Longleftrightarrow \quad B\,\underbrace{(-K_I e^i_\infty)}_{u_\infty} = (I - A)\,x_{\text{ref}}.$$

The final input $u_\infty = -K_I e^i_\infty$ is exactly the **holding input** (Concepts §7.4). The integrator has *learned* it from the accumulated error, whereas the prefilter of Task 2 *computed* it from the model.

Example (system a of Part F): $e^i_\infty = [-0.5,\ 0.5]^\top$, $u_\infty = -\begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}\begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$, and $Bu_\infty = [0.1,\ -0.1]^\top = (I - A)x_{\text{ref}}$ ✓.

E.3* Task 3.3 — When $\tilde{A}$ is only Lyapunov stable

"Lyapunov stable" means: all $|\lambda| \le 1$, at least one $|\lambda| = 1$, and the trajectories stay bounded. Then $\tilde{A}^k \not\to 0$, so Steps 1–2 of E.2 fail. There are two ways $|\lambda| = 1$ can happen.

Case 1: $\lambda = 1$ is an eigenvalue.

- $I - \tilde{A}$ is **singular**. The equilibrium equation $(I - \tilde{A})\bar{z} = \tilde{B}x_{\text{ref}}$ then has either **no solution** or **infinitely many**, and the geometric series $\sum_k \tilde{A}^k$ from Task 1 diverges. There is no unique limit to evaluate, so the argument "evaluate the integrator row at the equilibrium" collapses.
- *Example* (1-D plant $a = 0.5$, $b = 1$, $T_s = 1$, $k_p = 0.5$, $k_I = 0$):

$$\tilde{A} = \begin{bmatrix} a - bk_p & -bk_I \\ T_s & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \lambda = 0,\ 1.$$

  The integral gain is zero, so this is just the P-controller of Task 1. The state settles at $x_\infty = 0.5$, a 50 % error. Meanwhile, the integrator keeps adding the error: $e^i_{k+1} = e^i_k - 0.5$, reaching $e^i_{40} = -20.5$. **The error remains, and the integrator state grows without bound.** This is integrator windup.

- **This case is unavoidable when** $m < n$ (official solution, first bullet): K_I is $m \times n$ with $m < n$, so there is a $w \neq 0$ with $K_I w = 0$. Then

$$\tilde{A} \begin{bmatrix} 0 \\ w \end{bmatrix} = \begin{bmatrix} (A - BK_p) \cdot 0 - BK_I w \\ T_s \cdot 0 + w \end{bmatrix} = \begin{bmatrix} 0 \\ w \end{bmatrix},$$

so $\lambda = 1$ is an eigenvalue **for every choice of** K_p, K_I. The integrator component along w never feeds back into the input. It simply keeps summing up its share of the error. (`task3_integrator.py` confirms $\lambda = 1$ for random gains on the single-input DC motor.)

Case 2: a complex pair $\lambda = e^{\pm i\varphi} \neq 1$**.**

- $\tilde{A}^k$ rotates without shrinking, so the transient never dies out and the tracking error **oscillates forever**.
- *Example* (same plant, $k_p = 0.5$, $k_I = 1$):

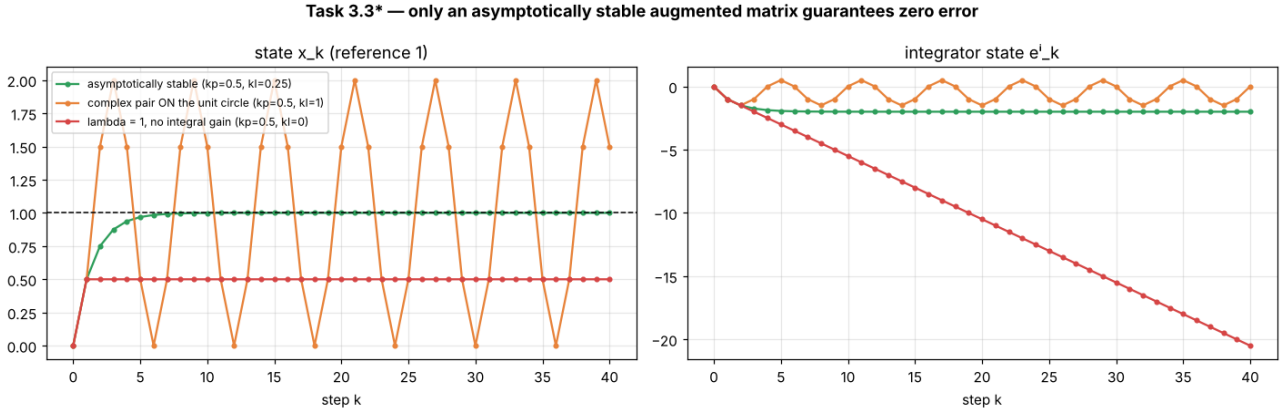
$$\tilde{A} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \qquad \det(\lambda I - \tilde{A}) = \lambda(\lambda - 1) + 1 = \lambda^2 - \lambda + 1 = 0,$$

$$\lambda = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2} = e^{\pm i\pi/3}, \qquad |\lambda| = 1.$$

Simulation from $x_0 = 0$, $e_0^i = 0$, $x_{\text{ref}} = 1$, using $x_{k+1} = -e_k^i + 0.5$ and $e_{k+1}^i = x_k + e_k^i - 1$:

k	0	1	2	3	4	5	6	7	...
x_k	0	0.5	1.5	2	1.5	0.5	0	0.5	repeats
e_k^i	0	−1	−1.5	−1	0	0.5	0	−1	repeats

The state swings between 0 and 2 with period 6 and never settles at 1 (`python/figures/task3_lyapunov_`



Practical implications for the tracking error.

1. **No guaranteed zero error:** depending on the case, a permanent offset remains (Case 1) or the error oscillates forever (Case 2).
2. **Integrator windup:** in Case 1 an integrator state grows without bound. In a real system the actuator saturates, and when the error finally changes sign the huge accumulated value causes a large overshoot that takes a long time to "unwind".
3. **Fragility:** eigenvalues *on* the unit circle are a boundary case. Any small model error can push them outside, making the loop unstable.

4. **Remedies:** design K_p, K_I so that all eigenvalues of $\tilde{A}$ lie strictly inside the unit circle, with some margin. With fewer inputs than states, integrate only m output errors that can actually be driven to zero (Part F, system c). Use anti-windup (limit or freeze the integrator when the actuator saturates).

   Answer to Task 3.3*. With $\tilde{A}$ only Lyapunov stable, $\tilde{A}^k \not\to 0$ and $I - \tilde{A}$ may be singular. The state need not converge, so $x_\infty = x_{\text{ref}}$ cannot be concluded. In practice the error either persists while the integrator winds up ($\lambda = 1$; always the case when $m < n$), or oscillates without decaying ($\lambda = e^{\pm i\varphi}$).

Part F — Numerical experiment: controller comparison

Systems (all with $n = 2$ states), step reference $x_{\text{ref}} = [1, 0]^\top$, and $x_0 = [0, 0]^\top$:

a) $A = \begin{bmatrix} 0.9 & 0.5 \\ 0.1 & 0.3 \end{bmatrix}$, b) $A = \begin{bmatrix} 1 & 0.5 \\ 0.1 & 0.7 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; c) $A = \begin{bmatrix} 0.9759 & 0.3452 \\ -0.005502 & -0.001944 \end{bmatrix}$, $B = \begin{bmatrix} 0.561 \\ 0.1304 \end{bmatrix}$

Assumption: the sampling time is not given on the sheet, so we use $T_s = 1$. T_s only appears as $K_I T_s$ (Concepts §7.8), so no conclusion depends on it. For the DC motor (c) the physical meaning of the states is in *system_models.pdf*, which is not available here; we simply call them x_1 and x_2.

Code: python/numerical_experiment.py. Every number below is printed by it.

F.1 Question 1 — Single- or multiple-input?

The number of inputs m is the number of **columns of** B.

system	size of B	m	rank B	type
a)	2×2	2	2	multiple-input
b)	2×2	2	2	multiple-input
c)	2×1	1	1	single-input

For a) and b), B is a permutation matrix: it swaps the two components, so $B^{-1} = B$ (check: $BB = I$). For c), one input drives both states.

F.2 Question 2 — Can the controllers of Tasks 1–3 achieve zero tracking error?

Step 1: eigenvalues of A (2×2 shortcut, Concepts §2.4):

- a) $\text{tr} = 1.2$, $\det = 0.27 - 0.05 = 0.22$: $\lambda = \frac{1.2 \pm \sqrt{1.44 - 0.88}}{2} = \frac{1.2 \pm 0.748}{2} = 0.974,\ 0.226$. Open loop stable.
- b) $\text{tr} = 1.7$, $\det = 0.7 - 0.05 = 0.65$: $\lambda = \frac{1.7 \pm \sqrt{2.89 - 2.6}}{2} = \frac{1.7 \pm 0.539}{2} = 1.119,\ 0.581$. **Open loop unstable.**
- c) $\text{tr} = 0.973956$, $\det = 0.9759(-0.001944) - 0.3452(-0.005502) = -0.0018971 + 0.0018993 = 2.14 \cdot 10^{-6}$: $\lambda \approx 0.974,\ 0.000002$. One slow mode, and one that dies out almost immediately.

Step 2: can K **be designed at all? (reachability)**

- a), b): B is invertible, so $W_c = [B, AB]$ already has rank 2. Reachable.

- c): $AB = [0.9759 \cdot 0.561 + 0.3452 \cdot 0.1304,\ -0.005502 \cdot 0.561 - 0.001944 \cdot 0.1304]^\top = [0.5925,\ -0.00334]^\top$, and $\det W_c = 0.561(-0.00334) - 0.5925 \cdot 0.1304 = -0.0791 \neq 0$. Reachable.

So in all three cases K can place the eigenvalues of $A - BK$ anywhere.

Step 3: can $x_{\text{ref}} = [1, 0]$ be held at all? (Concepts §7.4: $(I - A)x_{\text{ref}} \in \operatorname{im} B$?) $(I - A)x_{\text{ref}}$ is the first column of $I - A$:

- a) $[0.1,\ -0.1]^\top$. B is invertible, so yes, with $\bar{u} = B^{-1}[0.1, -0.1]^\top = [-0.1,\ 0.1]^\top$.
- b) $[0,\ -0.1]^\top$. Yes, with $\bar{u} = [-0.1,\ 0]^\top$.
- c) $[0.0241,\ 0.005502]^\top$. Is it $\bar{u}\,[0.561,\ 0.1304]^\top$ for some scalar $\bar{u}$? Row 1 gives $\bar{u} = 0.0241/0.561 = 0.04296$; row 2 gives $\bar{u} = 0.005502/0.1304 = 0.04219$. **Contradiction, so no.** All states that *can* be held lie on the line $\bar{x} = (I - A)^{-1}B\,\bar{u}$. With $\det(I - A) = 0.0241 \cdot 1.001944 + 0.3452 \cdot 0.005502 = 0.026046$:

$$(I-A)^{-1}B = \frac{1}{0.026046}\begin{bmatrix} 1.001944 & 0.3452 \\ -0.005502 & 0.0241 \end{bmatrix}\begin{bmatrix} 0.561 \\ 0.1304 \end{bmatrix} = \frac{1}{0.026046}\begin{bmatrix} 0.607105 \\ 0.000056 \end{bmatrix} = \begin{bmatrix} 23.309 \\ 0.00215 \end{bmatrix}.$$

  So every holdable state has $x_2 = \frac{0.00215}{23.309}x_1 = 9.23 \cdot 10^{-5}\,x_1$. The closest holdable state to $[1, 0]$ is $[1,\ 9.23 \cdot 10^{-5}]$, with $\bar{u} = 1/23.309 = 0.0429$. **No controller of any kind can reach $[1, 0]$ exactly**, but the unavoidable miss in x_2 is only $9 \cdot 10^{-5}$.

Step 4: each controller.

Task 1 (P): zero error $\iff$ $Ax_{\text{ref}} = x_{\text{ref}}$: - a) $A[1, 0]^\top = [0.9,\ 0.1]^\top \neq [1, 0]^\top$: **no**. - b) $A[1, 0]^\top = [1,\ 0.1]^\top \neq [1, 0]^\top$: **no** (although the first component matches, which will show in the simulation). - c) $A[1, 0]^\top = [0.9759,\ -0.0055]^\top$: **no**.

Task 2 (P + prefilter): - a), b): $\operatorname{rank} B = 2 = n$, so choose a stabilizing K with BK invertible and $F = (BK)^{-1}(I - A) + I$. **Yes, for every x_{ref}.** - c): $m = 1 < n = 2$, so BK has rank 1 and the general formula is impossible; zero error would also need x_{ref} holdable (Step 3). **No, not exactly**: x_1 can be made exact, and x_2 misses by $9 \cdot 10^{-5}$.

Task 3 (PI): we need $\tilde{A}$ asymptotically stable, which requires $\operatorname{rank} B = n$ (Concepts §7.7, PBH at $\lambda = 1$). - a), b): $\operatorname{rank} B = 2$. The script confirms that the augmented pair has $\operatorname{rank} W_c = 4$ of 4. **Yes, for every x_{ref}**, and robust to model errors. - c): $\operatorname{rank} B = 1$, so $\lambda = 1$ is always an eigenvalue of $\tilde{A}$ (Task 3.3*). The augmented pair has PBH rank 3 of 4 at $\lambda = 1$. Proposition 1.3 does not apply, and since $[1, 0]$ is not holdable, zero error is impossible anyway. **No.** Integrating only the x_1 error still makes x_1 exact.

	Task 1 (P)	Task 2 (P + F)	Task 3 (PI)
a)	no ($Ax_{\text{ref}} \neq x_{\text{ref}}$)	**yes**	**yes**
b)	no ($Ax_{\text{ref}} \neq x_{\text{ref}}$)	**yes**	**yes**
c)	no	only x_1 (x_2 off by $9 \cdot 10^{-5}$)	only x_1 ($\lambda = 1$ stuck; x_{ref} not holdable)

F.3 Question 3 — Designing K, F, K_p, K_I

System a) K (method 1 of Concepts §7.8): prescribe $A - BK = D := 0.5I$. Then $BK = A - 0.5I$, so $K = B^{-1}(A - 0.5I) = B(A - 0.5I)$. Multiplying by B from the left swaps the two rows:

$$A - 0.5I = \begin{bmatrix} 0.4 & 0.5 \\ 0.1 & -0.2 \end{bmatrix} \implies K = \begin{bmatrix} 0.1 & -0.2 \\ 0.4 & 0.5 \end{bmatrix}.$$

Check: BK swaps the rows back to $\begin{bmatrix} 0.4 & 0.5 \\ 0.1 & -0.2 \end{bmatrix}$, so $A - BK = 0.5I$, with eigenvalues $0.5,\ 0.5$. Stable ✓.

$F = (BK)^{-1}(I - A) + I$**.**

- $BK = \begin{bmatrix} 0.4 & 0.5 \\ 0.1 & -0.2 \end{bmatrix}$, $\det(BK) = 0.4(-0.2) - 0.5(0.1) = -0.13$.
- $(BK)^{-1} = \frac{1}{-0.13}\begin{bmatrix} -0.2 & -0.5 \\ -0.1 & 0.4 \end{bmatrix} = \frac{1}{13}\begin{bmatrix} 20 & 50 \\ 10 & -40 \end{bmatrix} \approx \begin{bmatrix} 1.5385 & 3.8462 \\ 0.7692 & -3.0769 \end{bmatrix}$.
- $I - A = \begin{bmatrix} 0.1 & -0.5 \\ -0.1 & 0.7 \end{bmatrix}$.
- $(BK)^{-1}(I - A) = \frac{1}{13}\begin{bmatrix} 20(0.1) + 50(-0.1) & 20(-0.5) + 50(0.7) \\ 10(0.1) - 40(-0.1) & 10(-0.5) - 40(0.7) \end{bmatrix} = \frac{1}{13}\begin{bmatrix} -3 & 25 \\ 5 & -33 \end{bmatrix}$.
- Add $I = \frac{1}{13}\begin{bmatrix} 13 & 0 \\ 0 & 13 \end{bmatrix}$:

$$F = \frac{1}{13}\begin{bmatrix} 10 & 25 \\ 5 & -20 \end{bmatrix} \approx \begin{bmatrix} 0.7692 & 1.9231 \\ 0.3846 & -1.5385 \end{bmatrix}.$$

Check: $BKF = (I - A) + BK = I - A + A - 0.5I = 0.5I$, so $A - BK + BKF = 0.5I + 0.5I = I$ ✓.

K_p, K_I **(method 3 of Concepts §7.8).** Target eigenvalues of $\tilde{A}$: $\lambda_1 = 0.5$, $\lambda_2 = 0.6$ (each twice). With $T_s = 1$:

$$\alpha = \lambda_1 + \lambda_2 - 1 = 0.1, \qquad \beta = \frac{\lambda_1\lambda_2 - \alpha}{T_s} = 0.3 - 0.1 = 0.2.$$

$$K_p = B^{-1}(A - 0.1I) = B\begin{bmatrix} 0.8 & 0.5 \\ 0.1 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.2 \\ 0.8 & 0.5 \end{bmatrix}, \qquad K_I = 0.2\,B^{-1} = \begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}.$$

Check: $A - BK_p = 0.1I$ and $BK_I = 0.2I$, so $\tilde{A} = \begin{bmatrix} 0.1I & -0.2I \\ I & I \end{bmatrix}$. By the block determinant rule (Concepts §7.6):

$$\det(\lambda I - \tilde{A}) = \big((\lambda - 0.1)(\lambda - 1) + 0.2\big)^2 = (\lambda^2 - 1.1\lambda + 0.3)^2 = \big((\lambda - 0.5)(\lambda - 0.6)\big)^2.$$

The eigenvalues are $0.5, 0.5, 0.6, 0.6$, all inside the unit circle ✓ (the script prints the same).

System b) Same B, same method.

K**:** $A - 0.5I = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.2 \end{bmatrix}$, so $K = \begin{bmatrix} 0.1 & 0.2 \\ 0.5 & 0.5 \end{bmatrix}$, and $A - BK = 0.5I$ ✓. The unstable open-loop eigenvalue 1.119 has been moved to 0.5.

F**:**

- $BK = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.2 \end{bmatrix}$, $\det = 0.1 - 0.05 = 0.05$, $(BK)^{-1} = 20\begin{bmatrix} 0.2 & -0.5 \\ -0.1 & 0.5 \end{bmatrix} = \begin{bmatrix} 4 & -10 \\ -2 & 10 \end{bmatrix}$.

- $I - A = \begin{bmatrix} 0 & -0.5 \\ -0.1 & 0.3 \end{bmatrix}$.
- $(BK)^{-1}(I-A) = \begin{bmatrix} 4(0)-10(-0.1) & 4(-0.5)-10(0.3) \\ -2(0)+10(-0.1) & -2(-0.5)+10(0.3) \end{bmatrix} = \begin{bmatrix} 1 & -5 \\ -1 & 4 \end{bmatrix}$.

$$F = \begin{bmatrix} 1 & -5 \\ -1 & 4 \end{bmatrix} + I = \begin{bmatrix} 2 & -5 \\ -1 & 5 \end{bmatrix}.$$

K_p, K_I ($\alpha = 0.1$, $\beta = 0.2$): $A - 0.1I = \begin{bmatrix} 0.9 & 0.5 \\ 0.1 & 0.6 \end{bmatrix}$, so

$$K_p = \begin{bmatrix} 0.1 & 0.6 \\ 0.9 & 0.5 \end{bmatrix}, \qquad K_I = \begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}, \qquad \text{eig}(\tilde{A}) = 0.5, 0.5, 0.6, 0.6 \checkmark.$$

System c) (single input) K **by coefficient matching** (method 2 of Concepts §7.8), with target eigenvalues 0.5 and 0.6, i.e. trace 1.1 and determinant 0.3. With $K = [k_1, k_2]$:

$$\text{tr}(A - BK) = \text{tr}\, A - b_1 k_1 - b_2 k_2 = 1.1 \implies 0.561\, k_1 + 0.1304\, k_2 = 0.973956 - 1.1 = -0.126044,$$

$$\det(A - BK) = \det A - k_1 \underbrace{(b_1 a_{22} - b_2 a_{12})}_{-0.046105} - k_2 \underbrace{(b_2 a_{11} - b_1 a_{21})}_{0.130344} = 0.3 \implies -0.046105\, k_1 + 0.130344\, k_2 =$$

(Coefficients: $b_1 a_{22} - b_2 a_{12} = 0.561(-0.001944) - 0.1304(0.3452) = -0.001091 - 0.045014$, and $b_2 a_{11} - b_1 a_{21} = 0.1304(0.9759) - 0.561(-0.005502) = 0.127257 + 0.003087$.)

Solve with Cramer's rule. The determinant of the coefficient matrix is $0.561(0.130344) - 0.1304(-0.046105) = 0.073123 + 0.006012 = 0.079135$.

$$k_1 = \frac{(-0.126044)(0.130344) - (0.1304)(-0.299998)}{0.079135} = \frac{-0.016429 + 0.039120}{0.079135} = 0.28673,$$

$$k_2 = \frac{(0.561)(-0.299998) - (-0.046105)(-0.126044)}{0.079135} = \frac{-0.168299 - 0.005811}{0.079135} = -2.20016.$$

$$K = [0.28673 \quad -2.20016], \qquad A - BK = \begin{bmatrix} 0.815043 & 1.579492 \\ -0.042892 & 0.284957 \end{bmatrix}.$$

Check: trace $= 1.1$ ✓, det $= 0.815043(0.284957) + 1.579492(0.042892) = 0.232252 + 0.067748 = 0.3000$ ✓, so the eigenvalues are 0.5 and 0.6. (`scipy.signal.place_poles` returns the same K.)

Task 1 prediction for c): $x_\infty = (I - M)^{-1} B\,(K x_{\text{ref}})$ with $K x_{\text{ref}} = k_1 = 0.28673$. Here $\det(I - M) = (1 - 0.5)(1 - 0.6) = 0.2$, and

$$g := (I - M)^{-1} B = \frac{1}{0.2} \begin{bmatrix} 0.715043 & 1.579492 \\ -0.042892 & 0.184957 \end{bmatrix} \begin{bmatrix} 0.561 \\ 0.1304 \end{bmatrix} = 5 \begin{bmatrix} 0.607105 \\ 0.000056 \end{bmatrix} = \begin{bmatrix} 3.03553 \\ 0.00028 \end{bmatrix},$$

$$x_\infty = 0.28673 \cdot \begin{bmatrix} 3.03553 \\ 0.00028 \end{bmatrix} = \begin{bmatrix} 0.87039 \\ 0.00008 \end{bmatrix}.$$

So the P controller only gets 87 % of the way in x_1.

F for c). Every steady state has the form $x_\infty = g \cdot (KFx_{\text{ref}})$, a multiple of the fixed vector g. Its direction cannot be changed; only its length can. Choose $F = fI$ so that the first component is exactly 1:

$$g_1 \cdot f\, k_1 = 1 \implies f = \frac{1}{g_1 k_1} = \frac{1}{3.03553 \cdot 0.28673} = \frac{0.329432}{0.28673} = 1.14892.$$

Then $x_\infty = [1,\ 0.00028/3.03553]^\top = [1,\ 9.2 \cdot 10^{-5}]^\top$, which is the closest holdable state found in F.2 Step 3. For a general reference $[r, 0]$ the same f gives $x_{1,\infty} = r$, because everything is linear.

K_p, K_I for c). A full-state integrator cannot work, because $\lambda = 1$ is stuck (F.2). Following the remedy of Task 3.3*, we **integrate only the x_1 error**: $K_I = [k_I,\ 0]$. The part that matters is a 3-state system with state (x, e_1^i):

$$\begin{bmatrix} x_{k+1} \\ e^i_{1,k+1} \end{bmatrix} = \begin{bmatrix} A & 0 \\ T_s[1\ 0] & 1 \end{bmatrix} \begin{bmatrix} x_k \\ e^i_{1,k} \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u_k + (\text{reference terms}),$$

whose state feedback $u = -[K_p\ k_I][e;\ e_1^i]$ is placed at eigenvalues $0.5, 0.6, 0.7$. This is the same coefficient-matching idea with 3 equations, solved by `place_poles`:

$$K_p = [0.64007 \quad -1.41966], \qquad K_I = [0.09883 \quad 0].$$

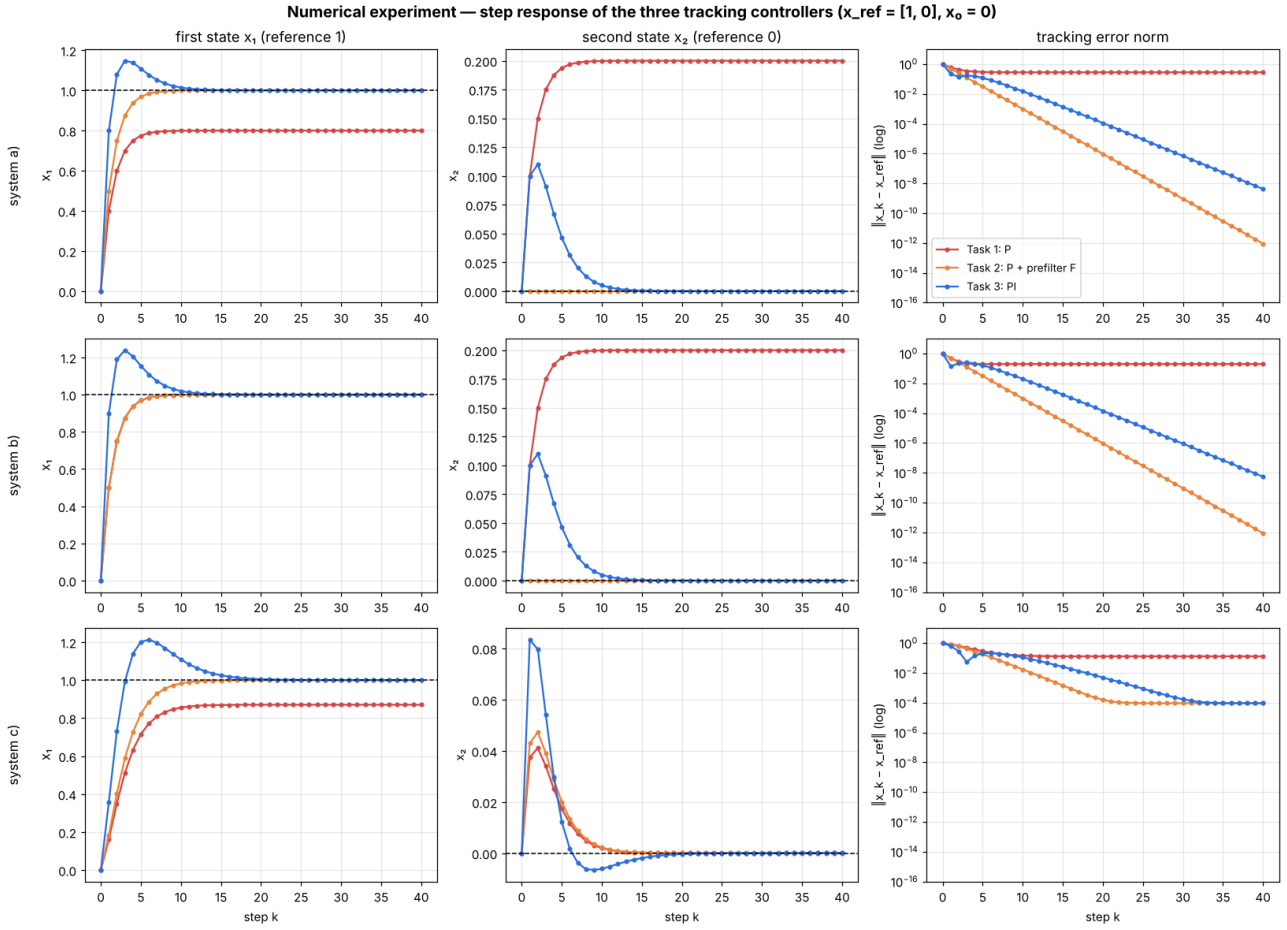
The full 4×4 $\tilde{A}$ then has eigenvalues $0.5,\ 0.6,\ 0.7,\ \mathbf{1}$. The 1 belongs to e_2^i, the direction in $\ker K_I$. That is exactly Task 3.3*.

F.4 Question 4 — Comparing the tracking errors

Simulation for 40 steps, with $x_{\text{ref}} = [1, 0]^\top$ and $x_0 = 0$:

system	controller	x_{40}	$\lVert x_{40} - x_{\text{ref}} \rVert$
a)	Task 1: P	$[0.8,\ 0.2]$	0.283
a)	Task 2: P + F	$[1,\ 0]$	$9 \cdot 10^{-13}$
a)	Task 3: PI	$[1,\ 0]$	$4 \cdot 10^{-9}$
b)	Task 1: P	$[1.0,\ 0.2]$	0.200
b)	Task 2: P + F	$[1,\ 0]$	$9 \cdot 10^{-13}$
b)	Task 3: PI	$[1,\ 0]$	$6 \cdot 10^{-9}$
c)	Task 1: P	$[0.87039,\ 0.00008]$	0.130
c)	Task 2: P + F	$[1,\ 0.00009]$	$9.2 \cdot 10^{-5}$
c)	Task 3: PI (x_1 integrated)	$[1,\ 0.00009]$	$9.2 \cdot 10^{-5}$

Hand check of the P results. With $A - BK = 0.5I$: $x_\infty = (0.5I)^{-1} BK x_{\text{ref}} = 2(A - 0.5I)x_{\text{ref}}$. - a) $2[0.4,\ 0.1]^\top = [0.8,\ 0.2]^\top$ ✓. - b) $2[0.5,\ 0.1]^\top = [1.0,\ 0.2]^\top$ ✓. The first component happens to be exact because the first component of Ax_{ref} equals 1.

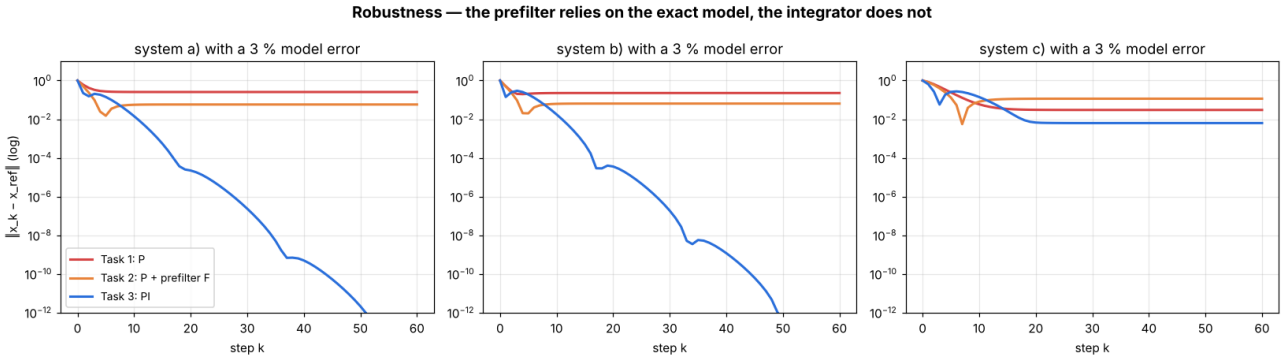


Discussion.

- **Task 1 (red)** settles quickly but at the wrong place, exactly as predicted by $(I - A + BK)^{-1}(A - I)x_{\text{ref}}$.
- **Task 2 (orange)** is the cleanest response for a) and b). With $A - BK = 0.5I$ and the prefilter, $x_{k+1} = 0.5x_k + 0.5x_{\text{ref}}$, so $x_k = (1 - 0.5^k)x_{\text{ref}}$: a straight path with no overshoot, x_2 stays exactly 0, and the error halves every step.
- **Task 3 (blue)** also reaches zero error, but more slowly (its slowest eigenvalue is 0.6 instead of 0.5), with overshoot (x_1 peaks at about 1.15 in a), 1.24 in b) and 1.21 in c)) and a detour in x_2. This is the typical price of integral action: the integrator must first build up the holding input.
- **System c)** confirms F.2: x_1 is tracked exactly by Tasks 2 and 3, and x_2 settles at $9.2 \cdot 10^{-5}$, the closest holdable state. No controller can do better.

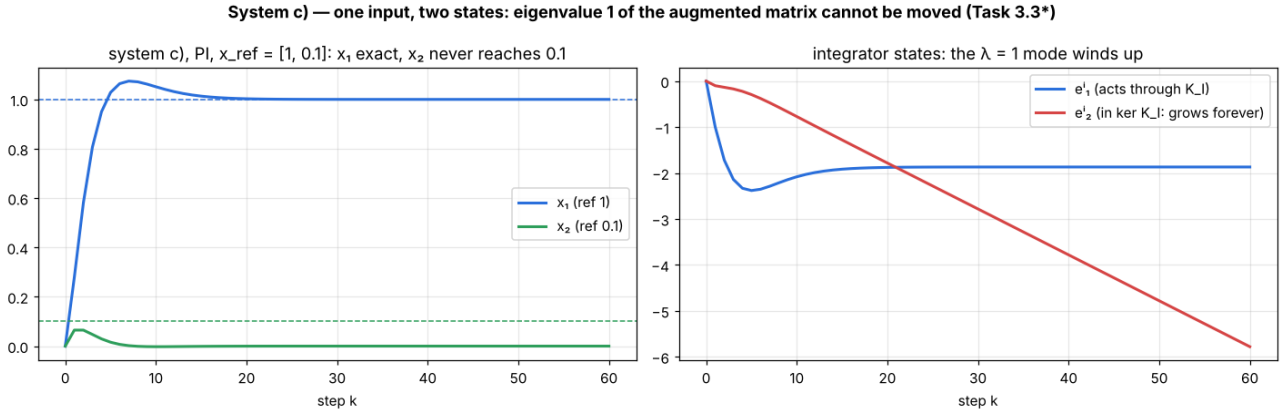
Extra 1: robustness to a 3 % model error (the true A is $1.03\times$ the design model, 60 steps):

	P	P + F	PI
a) final error	$[-0.148,\ 0.209]$	$[0.057,\ 0.006]$	$[0,\ 0]$
b) final error	$[0.071,\ 0.215]$	$[0.064,\ 0.007]$	$[0,\ 0]$
c) final error	$[-0.030,\ -0.006]$	$[0.115,\ -0.007]$	$[0,\ -0.006]$



The prefilter only removes the error for the exact model. With a model error the offset comes back, and in c) it is even worse than the plain P controller. The integrator keeps the error at zero for a) and b), and for x_1 in c), because it measures and corrects the error instead of predicting it (Part E.2, Step 5).

Extra 2: windup in system c). With an obviously unholdable reference $x_{\text{ref}} = [1,\ 0.1]$, the PI controller still brings x_1 to 1, but x_2 stays at $9 \cdot 10^{-5}$. The integrator component e_2^i (in $\ker K_I$, eigenvalue 1) grows by about -0.1 every step and reaches -5.8 after 60 steps, without bound. This is Task 3.3*, Case 1, in action.



Summary of the numerical experiment. - Multi-input systems a), b) (rank $B = n$): Task 1 leaves an error; Tasks 2 and 3 give zero error. The integrator is slower and overshoots, but it is robust to model errors. - Single-input DC motor c): no controller can hold $[1, 0]$ exactly, because it is not an equilibrium for any constant input. Tasks 2 and 3 make x_1 exact, and x_2 stays $9 \cdot 10^{-5}$ away. A full-state integrator always keeps an eigenvalue at 1.